ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. (CE/EE/ME/CSE)

Semester: II

SUBJECT CODE: BAS201

Physics

Time: 03:00 Hrs. **Maximum Marks: 70**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (14 Marks)

Q. 1. Answer the following questions: (1×7 = 7 Marks)

(i) What is the total internal reflection?
(ii) What is the role of chromium ions (Cr^{3+}) in a Ruby laser?
(iii) Define population inversion.
(iv) Write down the formula of specific rotation.
(v) What is a wave front?
(vi) What is Rayleigh's criterion for resolution?
(vii) What is the speed of light in vacuum?

Q. 2. Answer the following questions: (1×7 = 7 Marks)

(i) What is the path difference in constructive interference?
(ii) What is destructive interference?
(iii) What is stimulated emission?
(iv) How does a hologram differ from a normal photograph?
(v) What is the acceptance angle in optical fiber?
(vi) Write down the full form of LASER.
(vii) What is the main difference between E-ray and O-ray?

SECTION – B

NOTE:Attempt all questions. (7×2×2 = 28 Marks)

Q.3.(i)(a) Describe the Fraunhofer diffraction pattern due to a single-slit.

OR

(b) Find the thickness of a quarter wave plate when the wavelength of light is 5890Å. μ_e= 1.553 and μ_O= 1.544.

(ii)(a) Discuss about the construction, working, and energy level diagram of a Ruby laser.

OR

(b) Explain in detail the propagation mechanism in optical fiber with necessary diagrams.

Q.4.(i)(a) Calculate the resolving power of microscope if N.A. = 0.12 and the wavelength of light used is 6 X 10^{-5} cm.

OR

(b) Discuss the construction and working of a Helium-Neon laser with suitable figures. Give any 5 applications of lasers.

(ii) (a) Derive Lorentz transformation equations. Show that the four-dimensional volume element dx dy dz dt is invariant under Lorentz transformations.

OR

(b) Explain in detail the principle, formation, and theory of interference in a wedge-shaped film**.**

SECTION – C

NOTE: Attempt all questions. (14×2 = 28 Marks)

Q.5.(a) Suppose a thin transparent glass plate of thickness t is introduced in the path of one of the ray of a double slit experiment. Show that the central spot will be shifted by an amount given by: $\frac{D}{d}(\mu - 1)t$.

OR

(b) Explain the construction and working of a hologram with a neat diagram. What are the applications of holography?

Q.6.(a) Define Einstein's coefficients and find the relation between them. Give outlines of quantum theory to evaluate these coefficients.

OR

(b) Describe the production of plane-polarized, circularly and elliptically polarized light.